(I)

Let X be a topological space.

Consider outcome relations of the following nature:

$pRU,\ U \subseteq C \subseteq \overline{U},\ U \text{ open} \Rightarrow pRC.$

There will be need to work on subsets of X as we will see.

~~(II) Keeping (I) in mind, we will construct outcom~~

(II) It can be easily checked that GA need not be satisfied.

Recalling GA: $x \circ (y \circ z) = (x \circ y) \circ z$.

(III) Keeping (I) in mind, let us construct "suitable outcome relations on 'suitable space'.

Let $(D, \vee, \wedge)$ be a distributive lattice.

Then the set of prime filters of D will form a topological space whose basic closed sets are $\hat{a}$.

Recalling $\hat{a} = \{ p \mid a \in p \} \subseteq$ prime filters.

(IV) Let $(D, \vee, \wedge)$ be a distributive lattice and $\Diamond$ be a monotone map on D.

~~[illegible]~~ We already have definition of $Q_\Diamond$ given by Venema:

$pQ_\Diamond T$ iff $\Diamond a \in p$ for all $a \in F_T$.

Recalling $F_T := \{ a \in D \mid T \subseteq \hat{a} \}$.

Set X: prime filter space.
We will extract subsets of $Q_\Diamond$:

Define $Q'_\Diamond$ on the following ~~subspace~~ subspace of ~~X~~ X.

Y = open sets $\cup$ closed sets $\cup$ $\{c \subseteq X \mid U \subseteq c \subseteq \overline{U},\ U \text{ open}\}$

(i) $p Q'_\Diamond U$, U open iff $p Q_\Diamond U$.

(ii) $p Q'_\Diamond U$, U closed iff $p Q_\Diamond U$.

(iii) $p Q'_\Diamond C$, C neither closed nor open

iff $p Q_\Diamond U$ where $U \subseteq C \subseteq \overline{U}$ for some open set U.

Can be easily checked the 'defining feature' is satisfied.

(v) This relation $Q'_\Diamond$ need not be monotonic on Y, since something(s) like these may arise:

- $U \subseteq \overline{U} \subseteq C$, U open, C neither open nor closed

$p Q'_\Diamond U$ but $p \not{Q}'_\Diamond C$, meaning $p \not{Q}_\Diamond V$ for all open sets V s.t. $V \subseteq C \subseteq \overline{V}$

$\downarrow$

meaning

$p Q_\Diamond U$

More possibilities can lead to failure of total monotonicity. Above is just one example.
I am trying to come up with concrete counter-examples to solidify this.

(VI) Now let us try to work through the proof of composition:

Let $\mathcal{M} = (M, \vee, \wedge, -, \Diamond_g)_{g \in G}$ be a module over the game algebra $\mathcal{G} = (G, \vee, \wedge, -, \Diamond)$ and let g and h be arbitrary elements of G.

Then $Q'_{g \Diamond h} = Q'_g \circ Q'_h$.

Proof: Suppose $p(Q'_g \circ Q'_h)T$

$\therefore \exists U \in Y$, $pQ'_g U$ and $uQ'_h T$ for all $u \in U$.

Case 1: T is open/closed.

$uQ'_h T \Rightarrow uQ_h T$ (by definition of Q'_h).

Now consider an arbitrary element 'a' such that $T \subseteq \hat{a}$,

so, $\Diamond_h a \in u$ for all $u \in U$.

$\therefore \quad U \subseteq (\Diamond_h a)^{\wedge}$.

U open/closed : $p Q'_g U \Rightarrow p Q_g U$

$\therefore \Box_g \Box_h a \in p$

$\Box_g \Box_h a = \Box_{g \Delta h} a$ since M is a ~~model~~ module over the game algebra $\mathcal{G}$.

$\therefore \Box_{g \Delta h} a \in p$

U neither open, nor closed : $\exists$ V open s.t $V \subseteq U \subseteq \overline{V}$ and $p Q_g V$.

We have $U \subseteq (\Box_h a)^\wedge$

$\therefore V \subseteq (\Box_h a)^\wedge$

$\therefore \Box_g \Box_h a \in p$.

$\Box_g \Box_h a = \Box_{g \Delta h} a$.

$\therefore \Box_{g \Delta h} a \in p$.

Since $a \in F_T$ was arbitrary, we have $p Q'_{g \Delta h} T$.

Case 2: T is neither open, nor closed.

For all $u \in U$, $\exists\, T_u$ open s.t. $T_u \subseteq T \subseteq \overline{T_u}$ and

$$u \, Q_h \, T_u .$$

$$\therefore \; u \, Q_h (\cup T_u) \quad [\text{by monotonicity of } Q_h].$$

for all $u \in U$.

$\therefore \; u \, Q'_h (\cup T_u)$ for all $u \in U$ [since $\cup T_u$ is open].

By Case 1, $p \, Q'_{g \diamond h} (\cup T_u)$ [since $\cup T_u$ is open].

Now, $T_u \subseteq T \subseteq \overline{T_u}$

$$\Rightarrow \; \cup T_u \subseteq T \subseteq \cup \overline{T_u} \subseteq \overline{(\cup T_u)}$$

$\therefore \; p \, Q'_{g \diamond h} T .$

For the other direction, suppose $p\, Q'_{g \diamond h} T$.

<u>Case 1</u>: T is closed.

Then $p\, Q_{g \diamond h} T$.

We can copy Venema's proof to get a closed set U such that $p\, Q_g U$ and $u\, Q_h T$ for all $u \in U$.

The 'U' Venema used was this:

$$U = \bigcap \{ (\Diamond_h b)^\wedge \mid b \in F_T \}.$$

~~Proof~~ $\therefore$ $p\, Q'_g U$ and $u\, Q'_h T$ for all $u \in U$.

$\therefore$ $p\, (Q'_g \circ Q'_h)\, T$

<u>Case 2</u>: T is open.

~~Then $p\, Q_{g \diamond h} U$ and $p\, Q_{g \diamond h} U$~~

Then $p\, Q_{g \diamond h} T$ and $p\, Q_{g \diamond h} \overline{T}$ by monotonicity of $Q_{g \diamond h}$.

By Case 1, there is a closed set U such that

$p\, Q_g U$ and $u\, Q_h \overline{T}$ for all $u \in U$.

$u\, Q_h \overline{T} \implies u\, Q_h T$

$\therefore$ $p\, Q'_g U$ and $u\, Q'_h T$

$\therefore$ $p\, (Q'_g \circ Q'_h)\, T$

Case 3: T is neither open, nor closed.

$\therefore\ \exists\ V$ open, $V \subseteq T \subseteq \overline{V}$ and $p\, Q_{g \wedge h}\, V$.

$\therefore\ p\, Q_{g \wedge h}\, \overline{V}$ by Monotonicity of $Q_{g \wedge h}$.

By Case 1, $\exists$ closed set U st $p\, Q_g\, U$ and $u\, Q_h\, \overline{V}$.

~~$\therefore\ p\, Q'_g\, U$ and $u\, Q'_h$~~ .

$u\, Q_h\, \overline{V} \implies u\, Q_h\, V$.

$\therefore\ p\, Q'_g\, U$ and $u\, Q'_h\, V$. $T\ \forall\ u$

$\therefore\ p\, (Q'_g \circ Q'_h)\, T$.